

Sensor Scheduling in Electronic Support Using Markov Chains

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Abstract

In Electronic Support, receivers must maintain surveillance over the very wide portion of the electromagnetic spectrum in which threat emitters operate. A common approach is to use a receiver with a relatively narrow bandwidth which sweeps its centre frequency over the threat bandwidth to search for emitters. The sequence and timing of changes in the centre frequency constitute a search strategy. The search can be expedited if there is intelligence about the operational parameters of the emitters that are likely to be found. However, it can happen that the intelligence is deficient, untrustworthy or absent. In this case, what is the best search strategy to use? We propose a random search strategy based on a continuous-time Markov chain (CTMC). When the search is conducted for emitters with a periodic scan, we show that there is an optimal configuration for the CTMC. It is optimal in the sense that the expected time to intercept an emitter approaches linearity most quickly with respect to the emitter's scan period. A fast and smooth approach to linearity is important since other strategies can exhibit considerable and abrupt variations in the intercept time as a function of scan period. In theory and in numerical examples, we compare the optimum CTMC strategy with other strategies to demonstrate its superior properties.

I. INTRODUCTION

Receivers are a critical element of many surveillance systems that exist in modern military and civilian environments where their roles include target detection and identification, navigation and mapping, target tracking and weapons guidance. In Electronic Support, a key operational requirement is the ability to *detect* or *intercept* users of the electromagnetic spectrum, which we call emitters (usually radars), in the shortest possible time. However, the bandwidth in which emitters operate spans many gigahertz. In order to cope with this very large bandwidth, a widely favoured receiver architecture employs a receiver of more modest bandwidth with an agile centre frequency. The frequency-swept superheterodyne receiver is an example of this sort of receiver [1].

It is necessary to repeatedly re-tune the centre frequency of the receiver in order to maintain surveillance over the entire search bandwidth. The sequence and timing of changes to the centre frequency constitute a search strategy. This is a sensor scheduling problem. Typically, the search bandwidth is divided equally by the bandwidth of the receiver and the strategy involves tuning the receiver into each of the smaller bands in some sequence. In each band, we say the receiver *dwells* for a certain period before re-tuning to a different band.

An obvious and widely used strategy is the *periodic strategy*. Here, the receiver simply steps through each of the bands sequentially with (typically) equal dwell periods. Once the

receiver steps to the last band in the sequence, it begins again from the first. The time to complete a sequence we call the *sweep period*.

On the other hand, emitters also commonly employ a periodic search strategy of their own. In order to gain good angular resolution, a radar may have a highly directional antenna. The main beam is then scanned, either through mechanical movement of the antenna or, in more modern and sophisticated emitters, through electronic ‘beam steering’. In these cases, the times at which the main beam of the emitter is pointed towards the receiver is periodic and this period is its *scan period*.

In operation, the goal of the receiver is to try to detect radiation from the emitter in the shortest possible time, in order that the operator can be informed, the emitter identified and appropriate action taken. How long will it take our receiver to intercept the emitter? In order for this to happen, the receiver must be dwelling in the ‘right’ band, *i.e.*, the one on which the emitter operates, while the emitter is pointing in the ‘right’ direction, *i.e.*, the one in which the receiver lies. If both emitter and receiver are employing periodic strategies then this is a periodic pulse-train interception problem.

Determining the intercept time when both emitter and receiver are using a periodic strategy requires some application of elementary number theory. The problem was first studied by Richards [2], who examined the problem of intercept for two strictly periodic pulse trains in connection with an (unstated) problem in theoretical physics. He noticed that ‘with certain rational ratios of the periods, the events may “lock in step”’. Hence, in our emitter/receiver problem, there is a possibility that the receiver may never intercept the emitter if the receiver sweep period is poorly chosen, a situation which we call *synchronisation*. However, Richards admits that, despite deriving certain approximations for probability of intercept, ‘the original problem is not completely solved’.

Miller & Schwarz [3] showed how intercept time for certain ratios of the periods of the pulse trains could be obtained using linear congruence when the periods were assumed to be commensurate, *i.e.*, both were integer multiples of some common ‘base’ period. Their work was refined by Friedman [4].

However, a general formulation of the problem first appeared in the work of Kelly *et al.* [5], who describe what is essentially an algorithm for determining the intercept time. Clarkson *et al.* [6] made clear the links with elementary number theory. They showed that intercept time is a problem of Diophantine approximation and can be solved either through the application of Euclid’s algorithm or by examination of adjacent fractions in a Farey series [7]. Recently,

Clarkson [8] showed that if the sweep period of the receiver is varied while the proportion of time spent in each band is held constant, then there is usually only a finite number of sweep periods that can cause synchronisation with the emitter.

The potential for synchronisation in the periodic strategy is of concern. It is unsettling to think that, by improper choice of receiver sweep period, the receiver might never detect the emitter. Even if the sweep period lies near to one of the ‘synchronisation periods’, the intercept time may be arbitrarily long.

Can we avoid synchronisation? Washburn [9] suggested that one way to reduce the chances of synchronisation would be to introduce jitter into the receiver sweep period. Jitter can be best described as regulated noise in the sweep period of the receiver. This would have the effect of making the search pattern of the receiver ‘less’ periodic. Although a random search strategy of this type destroys any possibility of a guaranteed upper bound on the intercept time, it is possible to measure performance through, for instance, the expected intercept time. Kelly *et al.* [5] carried out extensive simulation studies to evaluate the effect of varying the amount of jitter. However, rigorous analysis of the strategy appears very difficult.

Another way to combat synchronisation is to have good intelligence about the scan periods of the emitters that are likely to be in operation. This is the approach of Clarkson [8], who describes a method for selecting a sweep period for a periodic receiver search strategy which minimises the maximum intercept time; the maximum being taken over all the emitters listed in a threat emitter list. When the intelligence is good, and the scan period parameters recorded in the threat emitter list can be relied upon, the sweep period setting calculated according to Clarkson’s method gives very low intercept times.

What should we do if we do not have good intelligence of emitter parameters? In this paper, we seek a search strategy that is, in some sense, maximally robust to uncertainties in the scan period of the emitters that might be encountered, whilst also being susceptible to analysis. We seek a strategy for which the intercept time varies smoothly as a function of emitter scan period. We propose a random strategy which is based on a continuous-time Markov chain (CTMC).

II. MATHEMATICAL BACKGROUND

A. Window Functions, Pulse Trains and the Interception Process

We model the process of detection or interception of an emitter by a receiver in terms of *window functions* or *pulse trains*. A window function is binary: ‘1’ or ‘0’, ‘on’ or ‘off’,

‘true’ or ‘false’. It is a function of time.

In the simplest mathematical formulation, interception is deemed to occur when two independent window functions are simultaneously ‘on’. One window function represents the times at which the receiver is dwelling on the band in which the emitter operates. The other window function represents the times at which the main beam of the emitter is illuminating the receiver. The situation is depicted in Figure 1. A more sophisticated model might also include a third window function representing individual RF pulses from the emitter, or the effect of scanning a directional antenna in the receiver, if such is being used.

Note also that some authors make a distinction between interception and detection: the former is said to occur when any energy at all is registered at the receiver, the latter occurs only when enough energy has been received in order to positively identify its source. We tend to use the terms interchangeably here.

B. Periodic Search Strategies

Both receiver and emitter may employ periodic search strategies. If the receiver moves from one band to the next, always staying for a fixed amount of time in each band then its window function is periodic. Similarly, if the main beam of an emitter is rotating at a constant speed, then its window function is also periodic.

Since periodic search strategies are an obvious choice for both emitters and receivers, it is instructive to briefly review the interception properties when both window functions are periodic. Let T_1 be the scan period of the emitter, *i.e.*, the time taken for an emitter to rotate 360° . Let T_2 be the sweep period of the receiver. This is the time taken by the receiver to sequentially search through all its bands. Further, let τ_1 be the illumination time of the emitter and let τ_2 be the dwell period of the receiver at the band on which the emitter is operating. We can also define a relative phase ϕ as the time difference between the midpoints of the first pulses from the two pulse trains. This is depicted in Figure 2.

Is there an upper bound on the intercept time, regardless of the relative phases between the two pulse trains? If so, we call this a *maximum* intercept time. On the other hand, we can ask if there is an expected intercept time, where the expectation is taken with respect to a probability distribution on the relative phase.

Following Clarkson [6], [8], we can calculate the expected time to intercept for a certain type of uniform distribution on the phases by first defining a PRI ratio $\alpha = T_2/T_1$ and the normalised sum of dwell times $\epsilon = (\tau_1 + \tau_2)/T_1$. If α is rational, so that we can write

$\alpha = h/k$, and $\epsilon < 1/k$ then the expected intercept time is infinite. In this case, there is synchronisation between emitter and receiver and, for certain phases, the two pulse trains are completely out-of-phase and intercept will never occur. When the pulse trains are not synchronised, we can find the intercept times by looking at adjacent elements in the Farey series of appropriate order.

The Farey series of order n is the series of fractions written in lowest terms in ascending order whose denominators are no greater than n [7]. It can be shown [10] that there is a unique pair of fractions h/k and h'/k' which are adjacent in a Farey series such that

$$|k\alpha - h| \leq \epsilon, \quad |k'\alpha - h'| \leq \epsilon, \quad |(k\alpha - h) - (k'\alpha - h')| > \epsilon.$$

Intercept is then guaranteed to occur within $k + k'$ sweeps of the receiver, *i.e.*, the maximum intercept time is $T_2(k + k')$.

In order to calculate the expected intercept time, we need to carefully define the distribution of the relative phases. We omit any detailed discussion here but note that, for a distribution which is uniform in a certain sense, the expected intercept time is infinite when the pulse trains are synchronised and, otherwise, is derived from a simple formula involving α , ϵ , k and k' [6].

C. A Generalised Matrix Inverse Based on Eigen Decomposition

An elementary fact from linear algebra is that not every square matrix has an inverse. However, it is possible to define a *generalised* or *pseudo* inverse which has some of the properties of the usual inverse. Specifically, a property of a (weak) generalised inverse $\mathbf{A}^-$ of a matrix $\mathbf{A}$ is that $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ and that $\mathbf{A}^-\mathbf{A}\mathbf{A}^- = \mathbf{A}^-$. A very famous example, which has additional desirable properties, is the MOORE-PENROSE pseudo inverse [11], [12], but it is by no means the only possible generalised inverse.

In our application, we will find that another type of generalised inverse is useful, based on the eigen decomposition or, more generally, the JORDAN decomposition of a matrix. By the Jordan decomposition theorem [13], we know that any square matrix $\mathbf{A}$ can be decomposed so that

$$\mathbf{A} = \mathbf{X}\mathbf{D}\mathbf{X}^{-1} \tag{1}$$

where, if $\mathbf{A}$ is non-singular, so too is $\mathbf{D}$, or otherwise $\mathbf{D}$ can be partitioned so that

$$\mathbf{D} = \begin{pmatrix} \Delta & 0 \\ 0 & 0 \end{pmatrix} \tag{2}$$

where Δ is non-singular. We define our generalised inverse $\mathbf{D}^e$ of $\mathbf{D}$ to be $\mathbf{D}^{-1}$ if it is non-singular or

$$\mathbf{D}^e = \begin{pmatrix} \Delta^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$$

otherwise. Finally, our generalised inverse $\mathbf{A}^e$ for an arbitrary square matrix $\mathbf{A}$ is

$$\mathbf{A}^e = \mathbf{X}\mathbf{D}^e\mathbf{X}^{-1}.$$

It is easily verified that the generalised inverse defined in this way meets the criteria for a weak generalised inverse. For non-defective matrices, it is clear that the generalised inverse defined in this way has the same eigenvectors as the original, but the non-zero eigenvalues are inverted. It is not in general equal to the MOORE-PENROSE pseudo inverse.

We will find the following lemma concerning this generalised inverse useful in the sequel. We use the notation $[\mathbf{x}]_i$ to denote the vector $\mathbf{x}$ from which the i^{th} element has been deleted and $[\mathbf{X}]_{i,j}$ to denote the matrix $\mathbf{X}$ from which the i^{th} row and j^{th} column have been deleted, i.e., a matrix minor.

Lemma 1: Let $\mathbf{A}$ be an $n \times n$ matrix which has a simple eigenvalue at 0 with associated left eigenvector $\mathbf{u}$ and right eigenvector $\mathbf{v}$. If we define the elements of a matrix $\mathbf{B}$ such that

$$b_{i,j} = v_i[\mathbf{u}]_i([\mathbf{A}]_{j,i})^{-1}[\mathbf{v}]_j u_j \quad (3)$$

then $\mathbf{B} = \mathbf{A}^e$.

Proof: We make the following observations.

- 1) We can rewrite (3) so that

$$b_{i,j} = v_i[\mathbf{u}^{(i)}]_n([\mathbf{A}^{(j,i)}]_{n,n})^{-1}[\mathbf{v}^{(j)}]_n u_j$$

where, for vectors, a superscript (k) indicates elementwise rotation of the elements by k positions to the left and, for matrices, a superscript (k,l) indicates that the rows are rotated upwards by k positions and the columns are rotated left by l positions.

- 2) With $\mathbf{A}$ decomposed as in (1) and writing $\mathbf{Y} = \mathbf{X}^{-1}$ for notational convenience, we observe that

$$\mathbf{A}^{(j,i)} = \mathbf{X}^{(j,0)}\mathbf{D}\mathbf{Y}^{(0,i)}$$

where $\mathbf{D}$ can be partitioned according to (2) and, in that partitioning, Δ is an $(n-1) \times (n-1)$ matrix.

3) Because the last row and column of $\mathbf{D}$ are all zero, it can be verified that

$$[\mathbf{A}^{(j,i)}]_{n,n} = [\mathbf{X}^{(j,0)}]_{n,n} \mathbf{\Delta} [\mathbf{Y}^{(0,i)}]_{n,n}.$$

4) It follows that

$$b_{i,j} = v_i [\mathbf{u}^{(i)}]_n ([\mathbf{Y}^{(0,i)}]_{n,n})^{-1} \mathbf{\Delta}^{-1} ([\mathbf{X}^{(j,0)}]_{n,n})^{-1} [\mathbf{v}^{(j)}]_n u_j. \quad (4)$$

5) We observe that $\mathbf{u}$ is the last row of $\mathbf{Y}$ and $\mathbf{v}$ is the last column of $\mathbf{X}$.

6) It follows that $\mathbf{u}^{(i)}$ is the last row of $\mathbf{Y}^{(0,i)}$ and $\mathbf{v}^{(j)}$ is the last column of $\mathbf{X}^{(j,0)}$.

7) Observe that $(\mathbf{X}^{(k,0)})^{-1} = \mathbf{Y}^{(0,k)}$ for any k .

8) Partitioning $\mathbf{X}^{(k,0)}$ and $\mathbf{Y}^{(0,k)}$ so that

$$\mathbf{X}^{(k,0)} = \begin{pmatrix} [\mathbf{X}^{(k,0)}]_{n,n} & [\mathbf{v}^{(k)}]_n \\ \boldsymbol{\xi}_k & v_k \end{pmatrix} \quad \text{and} \quad \mathbf{Y}^{(0,k)} = \begin{pmatrix} [\mathbf{Y}^{(0,k)}]_{n,n} & \boldsymbol{\eta}_k \\ [\mathbf{u}^{(k)}]_n & u_k \end{pmatrix}$$

it follows from the Block Matrix Inversion Lemma [14, Theorem 8.5.11], that

$$[\mathbf{X}^{(k,0)}]_{n,n} = ([\mathbf{Y}^{(0,k)}]_{n,n} - \frac{1}{u_k} \boldsymbol{\eta}_k [\mathbf{u}^{(k)}]_n)^{-1},$$

$$[\mathbf{Y}^{(0,k)}]_{n,n} = ([\mathbf{X}^{(k,0)}]_{n,n} - \frac{1}{v_k} [\mathbf{v}^{(k)}]_n \boldsymbol{\xi}_k)^{-1}.$$

9) Hence,

$$[\mathbf{u}^{(i)}]_n ([\mathbf{Y}^{(0,i)}]_{n,n})^{-1} = [\mathbf{u}^{(i)}]_n ([\mathbf{X}^{(i,0)}]_{n,n} - \frac{1}{v_i} [\mathbf{v}^{(i)}]_n \boldsymbol{\xi}_i) = -\frac{1}{v_i} \boldsymbol{\xi}_i,$$

$$([\mathbf{X}^{(j,0)}]_{n,n})^{-1} [\mathbf{v}^{(j)}]_n = ([\mathbf{Y}^{(0,j)}]_{n,n} - \frac{1}{u_j} \boldsymbol{\eta}_j [\mathbf{u}^{(j)}]_n) [\mathbf{v}^{(j)}]_n = -\frac{1}{u_j} \boldsymbol{\eta}_j.$$

10) Substituting into (4), we find that

$$b_{i,j} = \boldsymbol{\xi}_i \mathbf{\Delta}^{-1} \boldsymbol{\eta}_j.$$

11) This is equivalent to

$$b_{i,j} = \begin{pmatrix} \boldsymbol{\xi}_i & u_n \end{pmatrix} \mathbf{D}^e \begin{pmatrix} \boldsymbol{\eta}_j \\ v_n \end{pmatrix} \quad (5)$$

Since the left-hand vector in (5) is the i^{th} row of $\mathbf{X}$ and the right-hand vector is the j^{th} column of $\mathbf{X}^{-1}$, it follows that $\mathbf{B} = \mathbf{A}^e$, as required. ■

Remark 1: A very similar expression to (3) can be derived for the MOORE-PENROSE pseudo inverse. Unfortunately for the application in this paper, the $[\mathbf{u}]_i$ and $[\mathbf{v}]_j$ terms are then transposed and occupy opposite positions around $([\mathbf{A}]_{j,i})^{-1}$.

D. Markov Chains

Our approach in this paper is to propose a search strategy based on a continuous-time Markov chain which avoids the problems of synchronisation which are exhibited by periodic search strategies. A Markov chain is a *stochastic process with state space $\mathcal{S}$* , i.e., it is a sequence $\{X_t\}$ of random variables defined on some probability space and taking values in $\mathcal{S}$. The variable t represents time. For a *discrete-time Markov chain (DTMC)*, the time variable t takes on integer values, whereas in a *continuous-time Markov chain (CTMC)* it takes real values. We assume that the set $\mathcal{S}$ is finite, i.e., we consider only *finite* Markov chains. Without loss of generality, we label the elements numerically: $1, \dots, |\mathcal{S}|$. Following the expositions in [15], [16], [17], [18], [19], we briefly review the properties of discrete- and continuous-time Markov chains that are relevant here.

1) *Discrete-Time Markov Chains*: For a DTMC, the process obeys the *discrete-time Markov property* that

$$\Pr\{X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, \dots\} = \Pr\{X_{n+1} = x_{n+1} \mid X_n = x_n\}, \quad (6)$$

for all $x_i \in \mathcal{S}$ and all integer n . That is, given X_n , the next state X_{n+1} is conditionally independent of $X_{n-1}, X_{n-2}, \dots$. A DTMC is called *time invariant* if the RHS of (6) is independent of n . A (time-invariant) DTMC can be conveniently defined in terms of its transition matrix $\mathbf{P}$ with elements

$$p_{ij} = \Pr\{X_{n+1} = j \mid X_n = i\}.$$

Note that the column vector $\mathbf{1}$, all of whose elements are 1, is a right eigenvector of $\mathbf{P}$ corresponding to the eigenvalue 1.

Under certain conditions which need not be described here, a Markov chain may have a *stationary distribution π* , i.e., a distribution $\Pr\{X_n = i\} = \pi_i$ which is independent of n . Note that a stationary distribution, when one exists, is a left eigenvector of $\mathbf{P}$ corresponding to the eigenvalue 1.

Given that, at time n , the Markov chain is in state i , how much time elapses until the chain enters state j for the first time? This is the *first-passage time* between states i and j when $i \neq j$. When the mean first-passage times exist, computation of

$$\mathbf{m}_j = ([\mathbf{I} - \mathbf{P}]_{j,j})^{-1} \mathbf{1} \quad (7)$$

yields a vector $\mathbf{m}_j$ of the mean first-passage times to state j from state i in ascending order of index, excluding the mean first-passage time from state j . (The matrix $\mathbf{I}$ is the identity matrix.)

We are now able to state and prove a simple theorem regarding mean first-passage times which draws on the generalised inverse introduced in Section II-C.

Theorem 1: Consider a DTMC with transition matrix $\mathbf{P}$ having a unique stationary distribution π . If the state of the DTMC at a certain time is distributed according to π then the mean first-passage time μ_j to state j is

$$\mu_j = \frac{b_{j,j}}{\pi_j}$$

where $\mathbf{B} = (\mathbf{I} - \mathbf{P})^e$.

Proof: Since the DTMC has a unique stationary distribution, the transition matrix has a simple eigenvalue at 1 for which π is the left eigenvector and $\mathbf{1}$ the right. From (7), we can verify that the mean first-passage time is

$$\mu_j = [\pi]_j ([\mathbf{I} - \mathbf{P}]_{j,j})^{-1} \mathbf{1}$$

where the final $\mathbf{1}$ could instead be written $[\mathbf{1}]_j$. Application of Lemma 1 then furnishes the result directly. ■

2) *Continuous-Time Markov Chains:* For a CTMC, the process obeys the *continuous-time Markov property* that

$$\Pr\{X_t = x_t \mid X_{t_0} = x_{t_0}, X_{t_1} = x_{t_1}, \dots\} = \Pr\{X_t = x_t \mid X_{t_0} = x_{t_0}\}$$

for all real $t > t_0 > t_1 > \dots$ and for all $x_t, x_{t_i} \in \mathcal{S}$. A (time-invariant) CTMC can be defined in terms of its transition matrix $\mathbf{P}(t)$, which we note is now a function of time difference, where

$$p_{ij}(t) = \Pr\{X_{s+t} = j \mid X_s = i\}.$$

We consider only finite, *regular* CTMCs, for which

$$\lim_{t \rightarrow 0^+} \mathbf{P}(t) = \mathbf{I}.$$

Regular CTMCs exhibit only a finite number of state transitions in any finite time interval with probability 1. In this case, it turns out that the CTMC can be equivalently defined in terms of its *rate matrix* $\mathbf{Q}$ where

$$\mathbf{P}(t) = e^{\mathbf{Q}t}$$

A property of the rate matrix is that every row sums to zero, the diagonal elements are non-positive and the off-diagonal elements are non-negative. Therefore, $\mathbf{1}$ is a right eigenvector of $\mathbf{Q}$ corresponding to the eigenvalue 0.

When the CTMC visits a state, the length of time it remains there is called the *sojourn time*. The mean sojourn time in state i is $1/q_i$ where $q_i = -q_{ii}$.

The CTMC is called *irreducible* if, for all i, j , $p_{ij}(t) > 0$ when $t > 0$. For an irreducible CTMC, the zero eigenvalue of $\mathbf{Q}$ is simple and all other eigenvalues have negative real parts. Complex eigenvalues occur in conjugate pairs. For an irreducible CTMC there is a unique stationary distribution π which is the left eigenvector corresponding to the zero eigenvalue.

Sometimes it is convenient to partition the states of a CTMC into ‘super-states’ or *lumped* states and to then analyse the behaviour of the chain in terms of the lumped states. However, it is not necessarily true that the lumped process is itself a CTMC. Without loss of generality, we can assume that the lumped states $\mathcal{L}_1, \dots, \mathcal{L}_k$ are disjoint intervals whose union is $\mathcal{S}$. It is then possible to partition the rate matrix so that

$$\mathbf{Q} = \begin{pmatrix} \mathbf{Q}_{\mathcal{L}_1, \mathcal{L}_1} & \cdots & \mathbf{Q}_{\mathcal{L}_1, \mathcal{L}_k} \\ \vdots & \ddots & \vdots \\ \mathbf{Q}_{\mathcal{L}_k, \mathcal{L}_1} & \cdots & \mathbf{Q}_{\mathcal{L}_k, \mathcal{L}_k} \end{pmatrix}.$$

The CTMC is called *lumpable* if, for each $\mathbf{Q}_{\mathcal{L}_i, \mathcal{L}_j}$, its row sums are identical. If the CTMC is lumpable, the lumped process is also a CTMC. For more information, see [20].

It is possible to calculate mean first-passage times for CTMCs in a similar way to that for DTMCs. That is, when the means exist, they can be calculated as follows:

$$\mathbf{m}_j = -([\mathbf{Q}]_{j,j})^{-1} \mathbf{1}$$

where $\mathbf{m}_j$ has the same meaning as it did for DTMCs in (7). We are therefore able to state the following, analogous theorem.

Theorem 2: Consider a regular CTMC with rate matrix $\mathbf{Q}$ having a unique stationary distribution π . If the state of the CTMC at a certain time is distributed according to π then the mean first-passage time ν_j to state j is

$$\nu_j = \frac{b_{j,j}}{\pi_j} \quad (8)$$

where $\mathbf{B} = -\mathbf{Q}^e$.

III. A MARKOV-CHAIN-BASED RECEIVER SEARCH STRATEGY

We now propose a random strategy for controlling an ES receiver. In this strategy, the sequence and timing of visits to each band are determined by a regular CTMC. That is, each of the n bands is represented by a state in the CTMC. Such a strategy could be easily implemented in software.

Since we assume the number of bands in the receiver is finite, so is the CTMC. We also constrain the CTMC to be irreducible. This is equivalent to requiring that the stationary distribution has $0 < \pi_i < 1$ in every band/state i . In the sequel, we will use the words ‘band’ and ‘state’ interchangeably.

When the receiver is first switched on, we start it in a state at random according to the stationary distribution.

IV. ANALYSIS OF INTERCEPT TIME

Suppose an emitter with scan period τ begins operating, or comes into range, in band i . Without loss of generality, suppose that it first scans past the receiver at time 0. Let $T_i(\tau)$ be the time at which our CTMC-based search strategy first intercepts the radar’s scan; the intercept time. The intercept time is the first time after (or at) time 0 that the receiver visits band i when the radar is illuminating the receiver. The intercept time is a random variable that depends on the random search pattern of the receiver. We can therefore define an expected intercept time $E[T_i(\tau)]$. Moreover, the expected intercept time can be maximised over the pulse width of the emitter pulse train. We call this the *maximum expected intercept time* or *MEIT*, $f_i(\tau)$. The intercept time is always maximised when the pulse width is 0. Let $\mathbf{f}(\tau)$ be the vector of these MEITs in each band.

Theorem 3: The MEIT of the CTMC-based receiver search strategy is

$$f_i(\tau) = \frac{\tau \phi_{i,i}(\tau)}{\pi_i} \quad (9)$$

where $\Phi(\tau) = (\mathbf{I} - e^{\mathbf{Q}\tau})^e$ and $\boldsymbol{\pi}$ is the stationary distribution of the CTMC.

Proof: Consider the emitter scans (of zero pulse width) which we seek to intercept. They occur at times $k\tau$ for $k = 0, 1, \dots$. Consider now the state of the receiver at these times. The state is a DTMC induced by the sampling period τ . The probability transition matrix of this DTMC is

$$\mathbf{P}(\tau) = e^{\mathbf{Q}\tau}.$$

We can now see that the MEIT in band i is related by a factor τ to the mean first-passage time to state i in the induced DTMC. The induced DTMC has a unique stationary distribution π , just as for the CTMC it is induced from. The result then follows immediately from application of Theorem 1. ■

The above theorem yields closed-form expressions for the MEIT of the proposed CTMC-based receiver search strategy. This is in contrast to the only other random search strategy of which the authors are aware, the jittered periodic search, for which the known analysis is only approximate [9], [5].

We conclude this section by observing two limits. Both limits are obtained in a straightforward fashion by consideration of the definition of the generalised inverse in Section II-C. First, we observe that the MEIT, in proportion to the scan period, approaches a constant as the scan period increases, *i.e.*,

$$\lim_{\tau \rightarrow \infty} \frac{f_i(\tau)}{\tau} = \frac{1 - \pi_i}{\pi_i}. \quad (10)$$

Hence, as the sweep period grows, the MEIT approaches linearity. The slope of the asymptotic line is determined by the stationary probability of being ‘in band’.

Second, the limit in the opposite direction is

$$\lim_{\tau \rightarrow 0} f_i(\tau) = \nu_i$$

where ν_i is the mean first-passage time of the CTMC to state i as defined in (8) in Theorem 2. Intuitively, this is reasonable since we expect that, for emitters with very fast scans, the dominant consideration in intercept time will be the time taken for the search to first return to the emitter’s band.

V. OPTIMISATION OF THE SEARCH STRATEGY

We have observed that the MEIT approaches linearity as the scan period, τ , of the sought emitter approaches infinity. However, at intermediate values of τ , the presence of terms of the form $1/(1 - e^{\lambda_i \tau})$ in the intercept-time expression control the rate at which linearity is approached and, if any eigenvalues are complex, may induce local maxima in the MEIT along the way, especially when normalised against the scan period.

Local maxima in the intercept time can be interpreted as mild forms of synchronisation. This is because, in this case, certain scan periods take longer on average to be intercepted than nearby scan periods, suggesting that there is a tendency in the receiver to ‘lock in step’ with the emitter scan period, at least for a while.

We would like to know whether there is a particular choice of CTMC strategy among all such strategies that is, in some sense, the ‘most linear’ in intercept time. That is, we would like to find a CTMC strategy whose MEIT approaches linearity quickly with scan period, with few or no local maxima. In the case where we have little information about the scan periods of emitters in our environment, we would at least be assured that there are no emitter scan periods which could, on average, evade interception for relatively long periods of time, *i.e.*, that there are no scan periods which can be expected to synchronise with the receiver.

To formalise our criterion for optimisation, we define an *approach function*, $g_i(\tau)$, where

$$g_i(\tau) = \frac{\pi_i f_i(\tau)}{\tau} + \pi_i. \quad (11)$$

The approach function measures the MEIT relative to scan period and further normalised against its asymptotic slope. From (10), we see that

$$\lim_{\tau \rightarrow \infty} g_i(\tau) = 1 \quad (12)$$

for each band i .

Let γ be a real number for which there exists some $\rho > 0$ such that

$$\max_i \sup_{\tau > \rho} g_i(\tau) - \frac{1 - \pi_i}{1 - e^{\gamma\tau}} \leq \pi_i$$

That is, we want to find some real γ for which each approach function is eventually bounded above by the function $\pi_i + (1 - \pi_i)/(1 - e^{\gamma\tau})$. The smaller γ is, the more quickly our approach function approaches its limit. For the sake of simplicity, we will confine further analysis to the case where the rate matrix is diagonalisable. It can be shown in this case that the infimum of all such γ is simply the real part of the largest non-zero eigenvalue of $\mathbf{Q}$. To see this, observe that, with a diagonalisable rate matrix with eigenvalues λ_j , $j = 1, \dots, n$, the MEIT of Theorem 3 can be written in the form

$$f_i(\tau) = \frac{\tau}{\pi_i} \sum_{j=1}^{n-1} \frac{z_{i,j}}{1 - e^{\lambda_j \tau}}$$

where the $z_{i,j}$ are constants occurring in complex conjugate pairs wherever the λ_j do. The summation is to $n - 1$ rather than n because $\lambda_n = 0$ and observe that, because of (12),

$$\sum_{j=1}^{n-1} z_{i,j} = 1 - \pi_i.$$

From (11), it follows that we can write $g_i(\tau)$ in the form

$$g_i(\tau) = \pi_i + \sum_{j=1}^{n-1} \frac{z_{i,j}}{1 - e^{\lambda_j \tau}}.$$

Now,

$$g_i(\tau) = \pi_i + \sum_{j=1}^{n-1} z_{i,j} \sum_{k=0}^{\infty} e^{k\lambda_j\tau}$$

and the sum over k converges uniformly because the λ_j have negative real parts. We can exchange the order of summation because the sum in j is finite, so that

$$g_i(\tau) = \pi_i + \sum_{k=0}^{\infty} \sum_{j=1}^{n-1} z_{i,j} e^{k\lambda_j\tau}.$$

Now, if γ is greater than the real part of any λ_j then there exists some ρ such that, for all $\tau > \rho$,

$$\sum_{j=1}^{n-1} z_{i,j} e^{k\lambda_j\tau} < e^{k\gamma\tau} \sum_{j=1}^{n-1} z_{i,j} = (1 - \pi_i) e^{k\gamma\tau}.$$

This implies that

$$g_i(\tau) < \pi_i + \frac{1 - \pi_i}{1 - e^{\gamma\tau}}.$$

Conversely, if γ is less than the real part of one of the λ_j then there is no such ρ for which this is true. Our objective in optimisation is to minimise the infimum of γ over all CTMC strategies.

On the other hand, we might like to place some restriction on the rate at which the receiver jumps between bands. In reality, it is likely that the receiver takes a non-zero amount of time to switch bands, during which time it is not (reliably) receiving energy. We would therefore like to control the amount of time spent re-tuning in some way. Ideally, we would model this re-tuning time directly in our formulation of the problem, but it complicates the analysis. Instead, we apply a constraint on the harmonic mean, H , of the average sojourn (dwell) times in each band, *i.e.*, we constrain $H = -n/(\text{tr } \mathbf{Q})$. By judicious choice of H , we exert some control over the amount of time the receiver spends jumping between bands. Incidentally, the harmonic mean of sojourn times has long been used as a measure of the *mobility* of Markov chains in economics [21], [22], [23].

Therefore, our aim is to minimise the largest real part of the non-zero eigenvalues in the rate matrix while holding its trace constant. However, the trace of the rate matrix is equal to the sum of the eigenvalues. The solution to this problem — minimising the maximum eigenvalue (or real part thereof) while holding the sum of the eigenvalues constant — is simply to make all of the real parts of the eigenvalues equal.

An optimal search strategy, then, is one governed by a CTMC whose rate matrix has identical eigenvalues apart from the mandatory zero eigenvalue. Such a rate matrix is

$$\mathbf{Q}^* = \frac{n}{(n-1)H}(\mathbf{1}\pi - \mathbf{I}).$$

It can be verified that this matrix satisfies the requirements of being a rate matrix, such as having one zero eigenvalue with associated left eigenvector π and right eigenvector $\mathbf{1}$, negative diagonal elements and positive off-diagonal elements, *etc.* It is diagonalisable and has $\text{tr } \mathbf{Q}^* = -n/H$. The MEIT is given by the expression

$$f_i(\tau) = \frac{(1 - \pi_i)\tau}{\pi_i[1 - \exp(-nH^{-1}\tau/(n-1))]}$$

and

$$\lim_{\tau \rightarrow 0} f_i(\tau) = \frac{1}{n\pi_i}H(n-1)(1 - \pi_i).$$

Finally, we note that, for any band, our optimal CTMC is lumpable into two states, ‘in band’ and ‘out of band’.

VI. NUMERICAL EXAMPLES

We now provide two numerical examples. Firstly, we will compare our optimal CTMC strategy against another CTMC strategy chosen arbitrarily. Secondly, we will compare our optimal CTMC strategy with a periodic strategy.

As the basis of our first comparison, we set the harmonic mean of the sojourn times to be $H = 10$ ms, the number of bands to be 25, and the stationary probability in each band to be equal. Therefore, the MEIT will approach a value of 24 scan periods in each band when the scan period is large and, for the optimal CTMC, 230.4 ms when the scan period is small.

We choose the CTMC of the competing strategy to be the circulant matrix whose first row is $(-\alpha, \alpha, 0, \dots, 0)$. This CTMC steps sequentially between bands from the first to the last and then back to the first again. In order match the harmonic-mean constraint, we must set $\alpha = 100 \text{ s}^{-1}$.

In Figure 3, we plot the MEIT of the optimal CTMC-based search strategy against the competing, sequential strategy. Observe that the sequential strategy exhibits a greater degree of variation about the asymptotic line than the optimal strategy. In particular, there is a ‘bump’ at 250 ms, suggesting that there is weak synchronisation with those scan periods. This is entirely to be expected, since on average the sequential strategy dwells in each band for 10 ms before moving onto the next band. With 25 bands to cycle through, one complete

cycle takes on average 250 ms and, by the central limit theorem, the cycle time is roughly normally distributed about this value.

In contrast, the optimal strategy shows no hint of synchronisation at any scan period.

In Figure 4, we have compared our optimal Markov search strategy with a periodic search strategy, employing the results from [8]. In the periodic search strategy the dwell time in each band is set to 10 ms over 25 bands, as in the sequential CTMC strategy. This satisfies the harmonic mean constraint, in order to have a fair comparison.

Observe that the MEIT of the periodic search strategy exhibits wild variations, with regular excursions to infinity. The excursions to infinity occur at the synchronisation ratios, and there are many of them. On the other hand, the low points of its MEIT are significantly lower than for the optimal CTMC strategy. This suggests that, if we have good and accurate intelligence about particular emitters and their scan periods, a periodic search strategy will do considerably better than the best CTMC strategy on average. However, if no such intelligence is available, the CTMC strategy is clearly to be preferred.

VII. CONCLUSION

We have proposed a search strategy for a frequency-agile receiver for Electronic Support based on a continuous-time Markov chain (CTMC). Firstly, we were able to show that such a strategy is analytically tractable, unlike any previously proposed random strategy known to the authors. The analysis was facilitated by a new method of computing mean first-passage times using generalised inverses. Secondly, we were able to show that there exists a CTMC strategy which does not exhibit synchronisation when searching for periodically scanned radars. Our ‘ideal’ for a search strategy is one in which the expected intercept time quickly and smoothly approaches linearity as a function of the radar’s scan period. We derived an optimal CTMC strategy according to a certain mathematical formulation of this ideal. Finally, we compared the performance of the optimal CTMC strategy against an arbitrarily chosen CTMC strategy and a periodic strategy. We demonstrated the superior performance of the optimal CTMC with regards to the linearity ideal. We found that the optimal CTMC strategy is to be preferred over a periodic strategy where there is little or no intelligence about the scan periods of the radars to be encountered.

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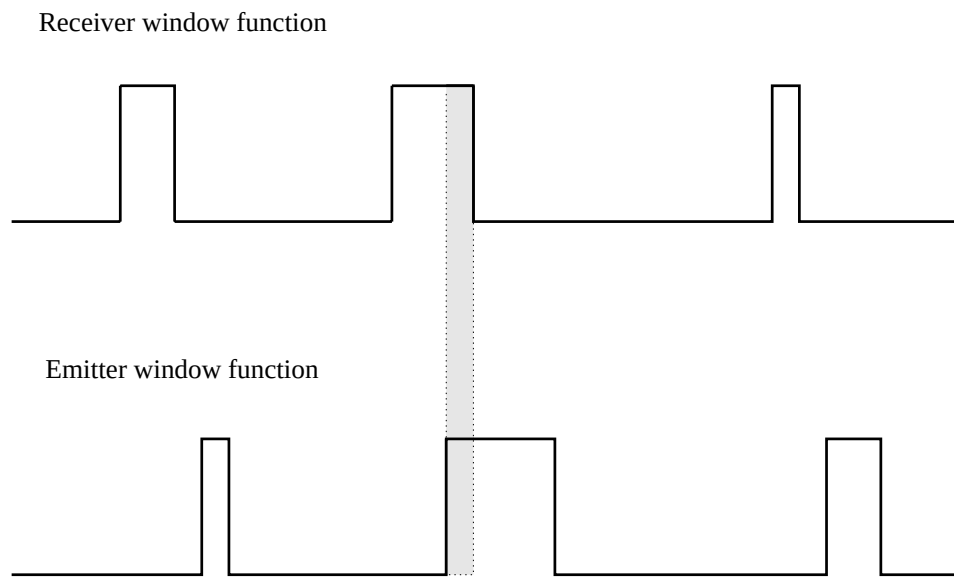


Fig. 1

RECEIVER DWELL AND EMITTER SCAN AS WINDOW FUNCTIONS.

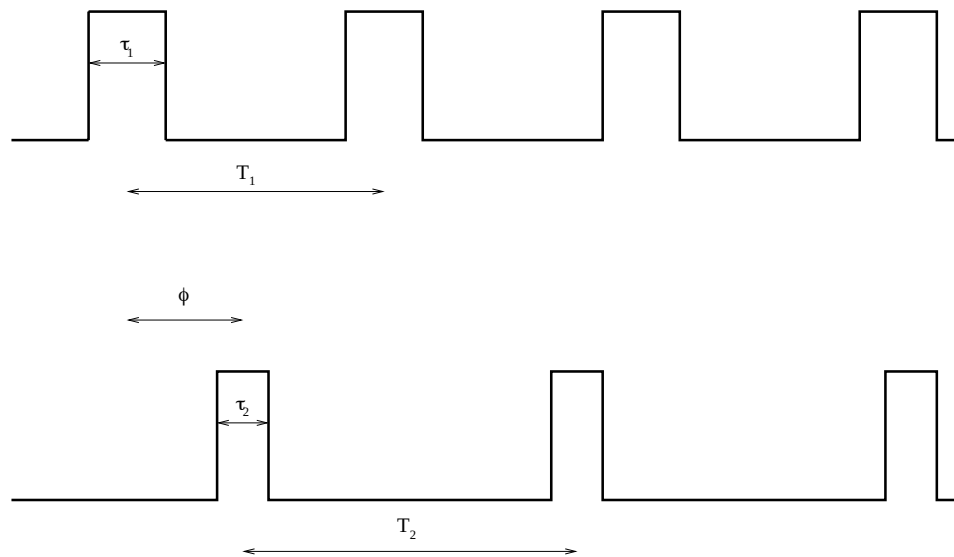


Fig. 2

EMITTER AND RECEIVER WINDOW FUNCTIONS FOR PERIODIC SEARCH STRATEGIES.

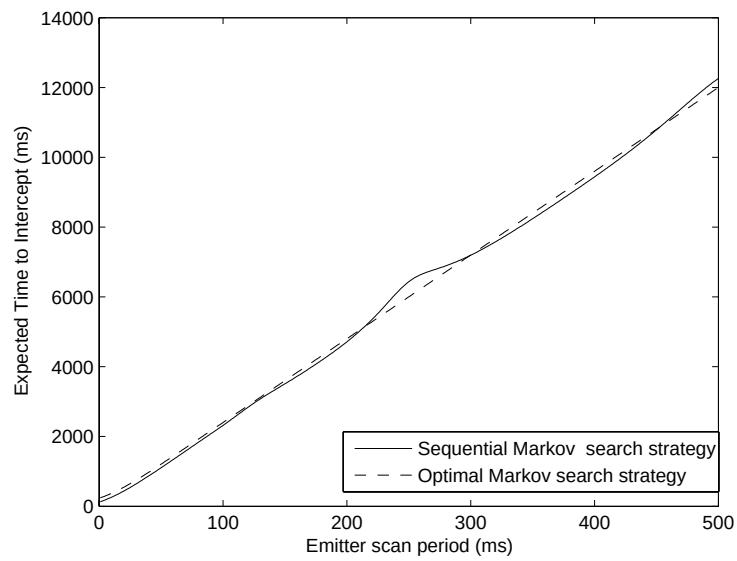


Fig. 3

COMPARISON OF THE INTERCEPT TIMES OF THE OPTIMAL CTMC-BASED SEARCH STRATEGY AGAINST A SEQUENTIAL CTMC-BASED STRATEGY.

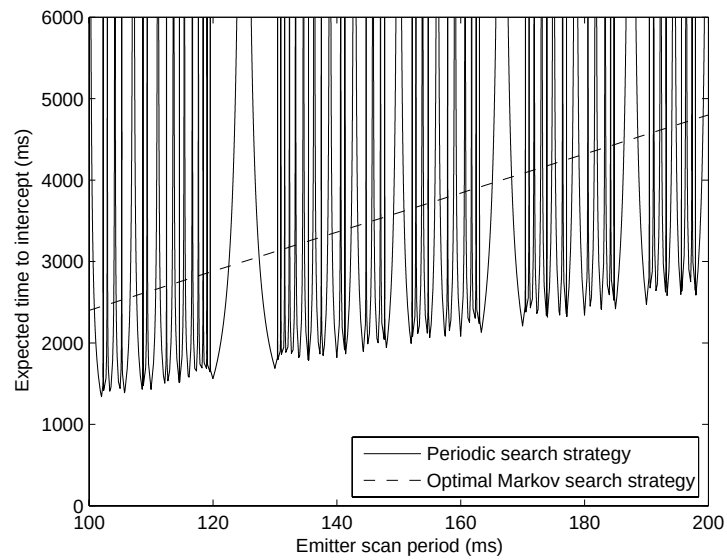


Fig. 4

COMPARISON OF THE INTERCEPT TIMES OF THE OPTIMAL CTMC-BASED SEARCH STRATEGY AGAINST A PERIODIC SEARCH STRATEGY.